Computational Methods in Relativistic Hydrodynamics
Applications to Astrophysical Mergers and Explosions

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- Relativistic hydrodynamics (RHD) plays a crucial role in modeling high-energy astrophysical phenomena.
- Examples include binary neutron star mergers[1].
- The thermodynamic properties of a relativistic gas are often modeled using the Synge equation of state[2].

[1] L. Baiotti and L. Rezzolla. "Binary neutron star mergers: a review of Einstein's richest laboratory". In: *Reports on Progress in Physics* 80.9 (2017), p. 096901

[2] J. L. Synge and P. M. Morse. *The relativistic gas*. 1958

Numerical Methods

Theorem Conservation Laws

The governing equations of relativistic hydrodynamics can be written as a system of hyperbolic conservation laws:

$$\frac{\partial \mathrm{U}}{\partial t} + \sum_{i=1}^{3} \frac{\partial \mathrm{F}^i(\mathrm{U})}{\partial x^i} = \mathrm{S}(\mathrm{U})$$

where U is the vector of conserved variables.

Definition Primitive Variables

The primitive variables (e.g., density ρ, velocity v, and pressure p) must be recovered from the conserved variables U at each time step, typically requiring numerical root-finding methods.

Simulation & Visualization

Dynamics of RHD Shockwaves

- Relativistic shockwaves exhibit complex interface instability.
- High-order WENO schemes are employed to resolve the shocks without spurious oscillations.
- The animation on the right shows the evolution of the density field over 21 frames.

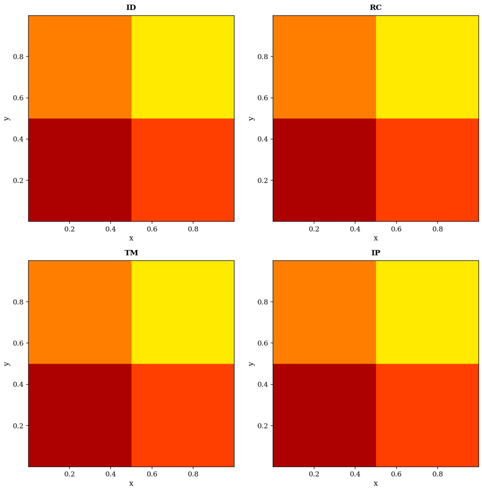

Figure: 2D Relativistic flow evolution.

- Robust numerical schemes are essential for simulating relativistic flows.
- The `mucomp` package provides support for standard theme styles, math blocks, footnoting/citations, and animations.